

THAARU Sepsis AI — Clinical Feature Engineering

Feature Derivation & Scaler Update Summary • Generated: July 19, 2026

1. Executive Summary

This phase derives clinically validated cardiovascular indices, patient-level temporal trajectory statistics, and measurement availability markers. To guarantee leak-proof scaling across splits, a new StandardScaler was fit exclusively on the Training split incorporating both original clinical columns and the new engineered features. The final dataset size increased from 68 columns to 97 total columns.

Feature Group	Features Added	Clinical Justification
Cardiovascular Indices	Shock Index, Pulse Pressure, MAP deviation	Identifiers for hemodynamic instability and hypovolemia.
Clinical Ratios	HR/MAP, Resp/O2Sat, HR/Temp	Proxy markers for physiological stress and SIRS indicators.
6h Rolling Stats	6h Mean, 6h Std/Variability (HR, Resp, Temp, MAP, O2Sat)	Captures vital signs instability and deterioration cycles.
1h Lag Differences	1h Slopes/Differences (HR, Temp, Resp, O2Sat)	Tracks acute worsening trends and rate of change.
Sparsity Markers	Hours since last Lactate, WBC, pH, Creatinine, etc.	Tracks clinician decision-making density (ordering patterns).
Timeline Flags	First 24 Hours, Diurnal proxy hour of stay	Controls for stay duration and temporal physiological cycle.

2. Processed Parquet Dimensions (Post Feature Engineering)

Dataset Partition	Rows (Hourly Records)	Original Columns	Engineered Columns	Final Columns
Training Split	1,086,436	68	29	97
Validation Split	231,472	68	29	97
Testing Split	234,302	68	29	97

3. Summary Statistics of Derived Clinical Features (Train Partition)

Below are the unscaled means, scales (stds), and variances fit on the training split, enabling stable and reproducible transformations during sequence building and model deployment.

Engineered Feature Name	Fit Mean	Fit Standard Deviation	Fit Variance
Shock_Index	0.7060	0.1990	0.0396
Pulse_Pressure	60.0088	19.5730	383.1012
MAP_deviation	-1.1233	7.2356	52.3541
HR_MAP_Ratio	1.0569	0.2851	0.0813
Resp_O2Sat_Ratio	0.1928	0.0516	0.0027
HR_Temp_Ratio	2.2858	0.4564	0.2083
HR_roll_mean_6h	84.3492	15.7513	248.1026
Resp_roll_mean_6h	18.6217	3.9957	15.9654
Temp_roll_mean_6h	36.8701	0.6087	0.3705
MAP_roll_mean_6h	82.4139	13.4285	180.3250
O2Sat_roll_mean_6h	97.2890	2.1905	4.7981
HR_roll_std_6h	5.3448	4.2740	18.2669
Resp_roll_std_6h	2.3981	1.7479	3.0551
Temp_roll_std_6h	0.2087	0.2123	0.0451
MAP_roll_std_6h	7.1337	4.7162	22.2427
O2Sat_roll_std_6h	1.2594	1.2105	1.4652
HR_diff_1h	0.0087	7.6761	58.9223
Temp_diff_1h	0.0028	0.2838	0.0806
Resp_diff_1h	0.0361	3.7439	14.0170
O2Sat_diff_1h	-0.0348	2.2094	4.8813
hours_since_last_Lactate	706.7053	449.1703	201753.9354
hours_since_last_Creatinine	220.1617	406.3733	165139.2294
hours_since_last_WBC	232.1916	414.1849	171549.1542
hours_since_last_Platelets	230.0279	412.7606	170371.3112
hours_since_last_BUN	211.6854	400.4394	160351.7196